



An improved asymmetric total synthesis of (+)-biotin via the enantioselective desymmetrization of a *meso*-cyclic anhydride mediated by cinchona alkaloid-based sulfonamide

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ARTICLE INFO

Article history:

Received 2 February 2010

Accepted 26 March 2010

Available online 4 May 2010

ABSTRACT

The highly enantioselective total synthesis of (+)-biotin **1** via the Hoffmann–Roche lactone–thiolactone strategy has been achieved starting from *cis*-1,3-dibenzyl-2-imidazolidone-4,5-dicarboxylic acid **2** with an overall yield of 35%. Two contiguous stereogenic centers at C-3a and C-6a were established through a rapid *cinchona* alkaloid-based sulfonamide-mediated enantioselective alcoholysis of *meso*-cyclic anhydride **3** to afford (4*S*,5*R*)-cinnamyl hemiester **4h**, the direct precursor to (3*S*,6*aR*)-lactone **5** with high enantioselectivity. A one-pot installation of the 4-carboxybutyl side chain was accomplished by a Fukuyama coupling reaction of (3*S*,6*aR*)-thiolactone **6** with the organozinc reagent prepared from ethyl 5-bromopentanoate.

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1. Introduction

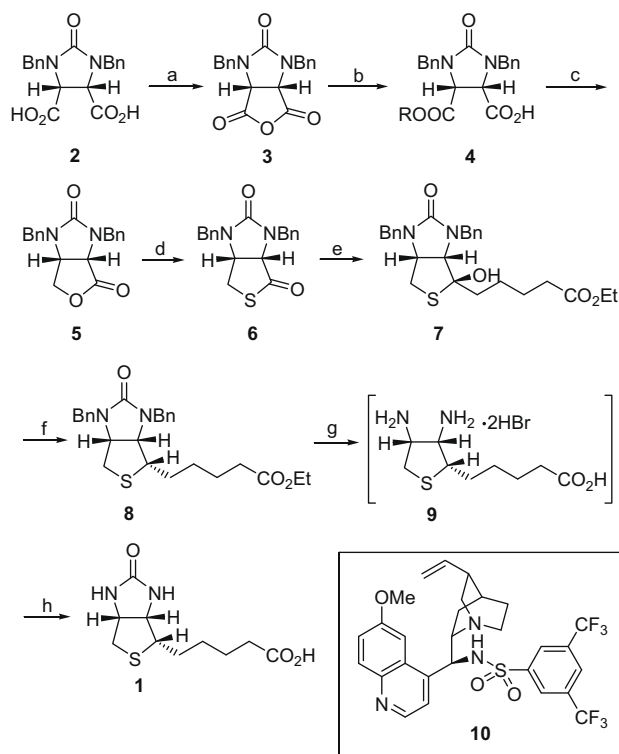
It is well known that the preparation of (3*S*,6*aR*)-lactone **5** from *meso*-cyclic acid **2** or its anhydride **3** is the most important step in the development of the commercial asymmetric total synthesis of (+)-biotin **1** via the Roche lactone–thiolactone strategy developed by Goldberg and Sternbach at the F. Hoffmann–La Roche company in 1949.¹ Not surprisingly, a considerable amount of effort devoted to this transformation in this process has led to tremendous achievements in the asymmetric desymmetrizations of **2** or **3** by utilizing different chiral techniques.² However, the chiral Lewis base-catalyzed enantioselective alcoholysis of **3** has been proven to be a robust and valid method for the preparation of the chiral (4*S*,5*R*)-hemiester or (4*R*,5*S*)-hemiester, the direct precursor to **5** in recent years.³ The first organocatalytic approach toward this key intermediate **5** described by Deng's group^{3a} in 2001 involved the bis-*cinchona* alkaloid-catalyzed enantioselective methanolysis of **3** and NaBH₄-mediated chemoselective reduction of the carboxyl group of (4*R*,5*S*)-hemiester after derivation with stoichiometric cyanuric chloride and *N*-methylmorpholine. The difficulties of scaling up **5** are well known in chemical process development due to the use of the expensive DHQD-PHN catalyst with relatively high molecular weight and the need for maintaining the cryogenic temperature (−40 °C) for 28 h in terms of the high enantioselectivity and yield. Two highly enantioselective conversions of **3** into the desired **4** using commercially available quinine^{3b} and its derivative

(*O*-propargylquinine)^{3c} as catalysts have been realized in our laboratory, but some limitations (low temperature, long reaction time) of Deng's approach have not been overcome. Our continued synthetic improvements for **4** have been carried out through desymmetrization of **3** catalyzed by the bifunctional thiourea-based catalysts derived from *cinchona* alkaloid and readily available (1*S*,2*S*)-2-amino-1-(*p*-nitrophenyl)propane-1,3-diol.^{3d} Although these two catalysts demonstrated excellent enantioselectivities and yields at room temperature, they suffer from self-aggregation, which results in low reactivity and strong dependence of the enantioselectivity on the reaction concentration and temperature as Song et al. have pointed out.⁴ Thus, the development of new versions of catalytic alcoholysis of **3** for the preparation of **4** to complete commercial asymmetric total synthesis of **1** is still in high demand.

Very recently, Song et al.⁵ reported that various enantiopure hemiesters could be efficiently prepared in excellent yields and enantiomeric purities via the bifunctional *cinchona* alkaloid-based sulfonamide-mediated alcoholysis of the corresponding *meso*-cyclic anhydrides without requiring extended reaction time at room temperature. This chiral technology offers a distinct advantage over the two most useful catalytic protocols for *meso*-cyclic anhydride desymmetrization with *cinchona* alkaloid derivatives as the general base catalysts developed by the groups led by Bolm⁶ and Deng^{3a} as well as recently reported bifunctional thiourea-based catalytic systems of Song himself and Cannon's group⁷ with respect to reaction rate, temperature, or/and concentration. Therefore, the promising work of Song's group came to our specific attention due to the parallels with our ongoing research concerning the asymmetric total synthesis of (+)-biotin

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Scheme 1. The synthetic route to (+)-biotin **1**. Reaction conditions: (a) toluene, reflux, 10 h, 99%; (b) ROH, MTBE, **10**, 25 °C; (c) KBH_4 , CaCl_2 , EtOH, 0–25 °C, then 5% HCl, 60 °C, 30 min, 98%; (d) EtSC(S)SNa, DMA, 125 °C, 5 h, 70%; (e) $\text{Br}(\text{CH}_2)_4\text{CO}_2\text{C}_2\text{H}_5$, Zn, DMF, THF, toluene, 10% Pd/C, 80–30 °C, 30 h, 75%; (f) NaBH_4 , TFA, DCM, 0–25 °C, 12 h, 91%; (g) 47% HBr, reflux, 48 h; (h) diphosgene, anisole, 40% KOH, 12 h, 82% (from **8** to **1**).

1 via a catalytic enantioselective alcoholysis strategy. Herein, we report an improved asymmetric total synthesis of **1** based upon the *cinchona* alkaloid-based sulfonamide-mediated enantioselective alcoholysis of *meso*-cyclic anhydride **3** into (4*S*,5*R*)-hemiester **4** and a one-pot introduction of a 4-carboxybutyl chain at C-4 in the (3*aS*,6*aR*)-thiolactone **6** by Fukuyama coupling reaction⁸ as key steps.

2. Results and discussion

Our asymmetric total synthesis of (+)-biotin **1** commenced with the commercially available *cis*-1,3-dibenzyl-2-imidazolidone-4,

5-dicarboxylic acid **2** (Scheme 1), which was subjected to dehydration in refluxing toluene with azeotropic removal of water to afford *meso*-cyclic anhydride **3** in almost quantitative yield and with 99% chemical purity.

With the prochiral cyclic anhydride **3** in hand, we were now in a position to study its desymmetrization into (4*S*,5*R*)-hemiester **4** to accomplish the asymmetric total synthesis of **1**. At the outset, we selected Song's protocol as being capable of facilitating the asymmetric methanolysis of various *meso*-cyclic anhydrides mediated by *cinchona* alkaloid-based sulfonamide **10**.⁵ However, under these conditions, the asymmetric desymmetrization of **3** using methanol as the test nucleophile in MTBE with 0.1 equiv of Song's catalyst **10** at ambient temperature affords the desired (4*S*,5*R*)-hemiester **4a** with very poor enantioselectivity (32% ee, Table 1, entry 1). However, the enantioselectivity of **4a** was improved up to 82% ee when 1.1 equiv of catalyst **10** was used (Table 1, entry 3). No improvement in the enantioselectivity was noticed on further increasing the catalyst loading from 1.1 to 1.2 equiv (Table 1, entry 4). The optimization of other reaction conditions (solvent, reaction temperature, and concentration) for this asymmetric methanolysis was also undertaken. It was found that amongst five different solvents examined (Table 1, entries 3, 11–14), the optimum solvent in terms of enantioselectivity was MTBE. The enantioselectivity of **4a** was also found to be strongly dependant upon the reaction temperature. As seen from entries 5–7 in Table 1, a significant decrease in enantioselectivities and reaction rates was observed when the reactions were conducted at 0 to –20 °C. We subsequently observed that the change of the reaction concentration of **3** in the solvent (from 0.15 to 0.025 M) led only to subtle variation in the enantioselectivities (Table 1, entries 3, 8–10).

The enantiopure methyl monoester **4a** (99% ee) could be easily obtained after a single recrystallization from EtOAc and its absolute configuration was established to be 4*S* and 5*R* by comparison with our previous reported values of the specific rotation.⁹ Furthermore, the *cinchona* alkaloid-based sulfonamide catalyst **10** was easily recovered quantitatively by the basification of its aqueous hydrochloride salt solution and could be reused without compromising either enantioselectivity or yield.

Having identified the optimized reaction conditions for the asymmetric desymmetrization of **3**, we next studied the effect of variation of the nucleophile structure on this alcoholysis. To this end, *meso*-cyclic anhydride **3** was treated with different alcohols in MTBE at room temperature in the presence of 1.1 equiv of catalyst **10**, and the results are summarized in Table 2. It is noteworthy that the ease of this asymmetric alcoholysis is dependant

Table 1
Optimization of the reaction conditions in the methanolytic desymmetrization of *meso*-cyclic anhydride **3**^a

Entry	Cat. (equiv)	Solvent	T (°C)	Concn ^b (M)	T (min)	Yield ^c (%)	ee ^d (%)
1	0.1	MTBE	25	0.1	60	99	32
2	0.8	MTBE	25	0.1	25	98	72
3	1.1	MTBE	25	0.1	5	99	82
4	1.2	MTBE	25	0.1	5	98	82
5	1.1	MTBE	0	0.1	30	99	72
6	1.1	MTBE	–10	0.1	60	97	64
7	1.1	MTBE	–20	0.1	120	98	54
8	1.1	MTBE	25	0.15	5	98	78
9	1.1	MTBE	25	0.05	5	99	82
10	1.1	MTBE	25	0.025	5	97	82
11	1.1	Et ₂ O	25	0.1	5	97	46
12	1.1	Dioxane	25	0.1	5	99	80
13	1.1	THF	25	0.1	5	98	78
14	1.1	Toluene	25	0.1	5	99	70

^a Unless otherwise noted, all reactions were performed with 0.5 mmol of **3** and 2.5 mmol of methanol.

^b Refers to the concentration of **3** in the solvent.

^c Yield of isolated product **4a**.

^d Determined by chiral HPLC analysis.

Table 2Asymmetric alcoholysis of *meso*-cyclic anhydride **3** with different alcohols^a

Entry	ROH	Product	Yield ^b (%)	ee ^c (%)
1	1-Propanol	4b	98	24
2	2-Propanol	4c	98	80
3	Allyl alcohol	4d	95	26
4	Propargyl alcohol	4e	97	24
5	Benzyl alcohol	4f	97	64
6	Cyclohexanol	4g	96	–6
7	<i>trans</i> -Cinnamyl alcohol	4h	98	92

^a Unless otherwise noted, all reactions were performed with 0.5 mmol of **3** and 2.5 mmol of ROH in 5 mL of MTBE at room temperature.^b Yield of isolated product.^c Determined by chiral HPLC analysis of the corresponding lactones, which were obtained by selective reduction of the ester group with LiEt₃H followed by acid-catalyzed lactonization.¹⁰

upon the steric properties of a nucleophile. The best result was achieved for the reaction of **3** with *trans*-cinnamyl alcohol yielding the corresponding chiral cinnamyl hemiester **4h** in excellent yield with enantioselectivities of 92% ee (Table 2, entry 7), whereas poor enantioselectivities (24% ee) were achieved when 1-propanol, allyl alcohol, and propargyl alcohol were employed as the nucleophiles (Table 2, entries 1, 3, and 4). In comparison with methanol, the ee of **4f** stemming from the ring-opening reaction of **3** with benzyl alcohol as the nucleophile was only moderate ee (64% ee) (Table 2, entry 5) and almost the racemic product **4g** was obtained when using cyclohexanol as the nucleophile (Table 2, entry 6).

The chemoselective reduction of ester function in **4h** was accomplished at room temperature using KBH₄ in the presence of anhydrous CaCl₂ in EtOH, followed by a lactonization on heating the resulting calcium salts of (4*S*,5*R*)-hydroxy acid with 5% aq HCl for 30 min to provide the desired key intermediate chiral lactone **5** in 98% overall yield with 99% enantiomeric excesses. Thiolactonization of **5** using sodium ethyl thioxanthogenate [EtSc(S)SNa] in anhydrous *N,N*-dimethylacetamide (DMA) at 125 °C for 5 h worked perfectly well to furnish a 70% yield of the chiral thiolactone **6**. The 10% Pd/C-catalyzed Fukuyama coupling of **6** with 5-ethoxy-5-oxopentylzinc bromide, in situ prepared from ethyl 5-bromopentanoate and zinc powder, smoothly proceeded in a mixed solvents of DMF, THF, and toluene at 30 °C for 30 h to afford the desired coupling product **7** in 75% yield.

In our earlier reported procedures,^{3b,c} the tertiary hydroxyl group of **7** was effectively reduced into the corresponding pentanoate **8** in excellent chemical yields upon treatment with triethylsilane and TFA or BF₃·OEt₂ in CH₂Cl₂ under mild reaction conditions. But these two methods required expensive triethylsilane, making them unacceptable in industrial synthesis. However, the ionic hydrogenation of **7** was realized at room temperature by using NaBH₄ as the reducing reagent in the presence of TFA in dichloromethane to afford the desired **8** in 91% yield. Its absolute stereochemistry at C-4 was confirmed by comparison with the reported values of the specific rotation.^{3c} The reduction did not go to completion when trichloroacetic acid or concd H₂SO₄ was utilized in the place of TFA as acid media even prolonging the reaction time from 12 to 24 h and enhancing the reaction temperature from room temperature to 60 °C.

One-pot debenzoylation and ring-opening which afford the diamine-2HBr salt **9** were carried out by heating the pentanoate **8** with 47% aq HBr at refluxing temperature for 48 h, which, without further purification, was allowed to react with trichloromethyl chloroformate in the presence of 40% KOH in anisole at 35 °C to provide (+)-biotin **1** in 82% overall yield.

3. Conclusions

In conclusion, we have successfully developed an improved synthetic process for the enantioselective total synthesis of (+)-biotin. The application of the *cinchona* alkaloid-based sulfonamide-mediated alcoholysis strategy for the construction of the two contiguous stereogenic centers of the biotin skeleton allows rapid access to (4*S*,5*R*)-cinnamyl hemiester **4h** in an enantiomerically enriched form under extremely mild conditions, which appears to be more compatible with industrial scale and has some advantages over the existing synthesis. The reactions described here could be an attractive alternative for the industrially viable synthesis of (+)-biotin.

4. Experimental

4.1. General

¹H and ¹³C NMR spectra were recorded on a Bruker Avance 400 spectrometer (400, 100 MHz, respectively) in CDCl₃ or DMSO-*d*₆ using tetramethylsilane (TMS) or DMSO (¹H δ 2.49) and CDCl₃ (¹³C δ 77.0) or DMSO-*d*₆ (¹³C δ 39.5) as internal standards. IR spectra were recorded on a JASCO FT/IR-4200 spectrometer. Mass spectra were recorded on a Waters Quattro Micromass instrument using electrospray ionization (ESI) techniques. Optical rotations were obtained on a JASCO P1020 digital polarimeter. Melting points were measured with a WRS-1B digital melting point apparatus and are uncorrected. HPLC analysis of the enantiomeric excesses of **4a** and **4b–4h** was performed using chiralcel OJ-H column (hexane/IPA/TFA = 92/8/0.2, λ = 220 nm, 0.5 mL/min) and OD-H column (hexane/IPA = 70/30, λ = 220 nm, 0.5 mL/min) Unless otherwise noted all reactions were conducted in oven-dried glassware under inert atmosphere of dried Ar or N₂. THF was distilled from sodium/benzophenone, toluene, and CH₂Cl₂ from calcium hydride and DMA from calcium hydride under reduced pressure. The alcohols used were purified according to standard methods,¹¹ other reagents were obtained from commercial sources and used as received.

4.2. *cis*-1,3-Dibenzyltetrahydro-2*H*-furo[3,4-*d*]imidazole-2,4,6-trione **3**

A mixture of **2** (10 g, 28 mmol) and toluene (190 mL) was stirred under reflux with azeotropic removal of water for 10 h. After cooled to room temperature, the precipitated colorless crystals were filtered and dried at 80 °C in vacuo for 3 h to give the pure product **3** as a white powder; yield: 9.31 g (99%); mp 237.1–237.6 °C [lit.¹² mp 237–239 °C]; IR (KBr): ν = 1805, 1740, 1687, 1227 cm^{–1}; ¹H NMR (CDCl₃): δ: 4.15 (s, 2H), 4.18 (s, 2H), 4.22 (d, 1H, *J* = 14.8 Hz), 5.15 (d, 1H, *J* = 14.8 Hz), 7.15–7.35 (m, 10H); MS (ESI): *m/z* = 337.4 (M+1)⁺.

4.3. General procedure for the asymmetric alcoholysis of **3** with various alcohols

Alcohol (2.5 mmol) was added dropwise into a suspension of **3** (0.168 g, 0.5 mmol) and **10** (0.329 g, 0.55 mmol) in MTBE (5 mL) under argon atmosphere. Upon completion of the addition the reaction mixture was stirred for 5 min at room temperature and then evaporated in vacuo. The residue was dissolved in AcOEt (10 mL), the solution was washed with 2 M HCl (3 × 10 mL), and dried over Na₂SO₄. Evaporation of the solvent yielded the corresponding hemiester **4a–4h**. To recover the *cinchona* alkaloid-based sulfonamide **10**, the combined acidic aqueous phases were basified with NH₄OH and extracted several times with CH₂Cl₂. After drying

over Na₂SO₄, the organic phase was concentrated in vacuo to give catalyst **10** in almost quantitative yield.

4.3.1. (4*S*,5*R*)-1,3-Dibenzyl-5-(methoxycarbonyl)-2-oxo-imidazolidine-4-carboxylic acid **4a**

White solid; yield: 0.182 g (99%); mp 145–147 °C; [α]_D^{25.2} = +6.0 (c 1.0, DMF) [lit.⁹ mp 150–151 °C; [α]_D²⁵ = +7.3 (c 1.0, DMF)]; ee = 82%; IR (KBr): ν = 3258, 2943, 1756, 1713, 1449, 1411, 1252, 968, 799, 598, 458 cm⁻¹; ¹H NMR (CDCl₃): δ : 3.63 (s, 3H), 4.13–4.04 (m, 4H), 4.98 (d, 1H, *J* = 14.8 Hz), 5.09 (d, 1H, *J* = 14.8 Hz), 6.85 (br s, 1H), 7.36–7.19 (m, 10H); ¹³C NMR (CDCl₃): δ : 46.80, 46.89, 52.6, 56.7, 57.3, 127.93, 127.98, 128.56, 128.64, 128.79, 128.85, 135.52, 135.57, 159.5, 168.5, 171.7; MS (ESI): *m/z* = 391.1 (M+Na)⁺.

4.3.2. (4*S*,5*R*)-1,3-Dibenzyl-5-(*n*-propoxycarbonyl)-2-oxo-imidazolidine-4-carboxylic acid **4b**

White solid; yield: 0.194 g (98%); mp 77–82 °C; [α]_D^{25.2} = +1.0 (c 1.0, CHCl₃) [lit.^{3b} mp 78–82 °C; [α]_D²⁵ = +2.5 (c 1.0, CHCl₃)]; ee = 24%; IR (KBr): ν = 3065, 2984, 1744, 1655, 1469, 1359, 1209, 1073, 952, 750, 700 cm⁻¹; ¹H NMR (CDCl₃): δ : 0.87 (t, 3H, *J* = 7.2 Hz), 1.58 (tq, 2H, *J* = 7.2, 7.2 Hz), 4.10–3.92 (m, 6H), 5.01 (d, 1H, *J* = 14.8 Hz), 5.08 (d, 1H, *J* = 14.8 Hz), 7.31–7.20 (m, 10H), 9.75 (br s, 1H); ¹³C NMR (CDCl₃): δ : 10.3, 21.6, 46.7, 46.8, 56.8, 57.3, 67.6, 127.94, 127.95, 128.59, 128.63, 128.80, 128.84, 135.59, 135.61, 159.7, 168.1, 171.8; MS (ESI): *m/z* = 419.1 (M+Na)⁺.

4.3.3. (4*S*,5*R*)-1,3-Dibenzyl-5-(isopropoxycarbonyl)-2-oxo-imidazolidine-4-carboxylic acid **4c**

Yellow oil; yield: 0.194 g (98%); [α]_D^{20.8} = –8.2 (c 1.0, DMF) [lit.¹³ [α]_D²⁰ = –10.1 (c 1.0, DMF)]; ee = 80%; IR (KBr): ν = 3438, 3028, 2979, 2494, 1956, 1697, 1446, 1235, 1108, 957, 836, 753, 702, 458 cm⁻¹; ¹H NMR (CDCl₃): δ : 1.04 (d, 6H, *J* = 6.0 Hz), 3.50 (m, 1H), 3.69–4.07 (m, 4H), 4.77 (d, 1H, *J* = 14.4 Hz), 4.79 (d, 1H, *J* = 14.4 Hz), 7.18–7.32 (m, 10H); ¹³C NMR (CDCl₃): δ : 8.8, 21.7, 25.9, 39.4, 46.3, 52.1, 58.2, 62.5, 68.7, 127.6, 137.2, 160.2, 168.8; MS (ESI): *m/z* = 419 (M+Na)⁺.

4.3.4. (4*S*,5*R*)-1,3-Dibenzyl-5-(allyloxycarbonyl)-2-oxo-imidazolidine-4-carboxylic acid **4d**

White solid; yield: 0.187 g (95%); mp 106–109 °C; [α]_D^{25.5} = +2.6 (c 1.0, CHCl₃) [lit.^{3b} mp 107.7–109.8 °C; [α]_D²⁵ = +7.8 (c 1.0, CHCl₃)]; ee = 26%; IR (KBr): ν = 3061, 2929, 1755, 1657, 1449, 1356, 1235, 1200, 936, 754, 703 cm⁻¹; ¹H NMR (CDCl₃): δ : 4.12–4.03 (m, 4H), 4.54–4.43 (m, 2H), 4.98 (d, 1H, *J* = 14.8 Hz), 5.06 (d, 1H, *J* = 14.8 Hz), 5.26–5.18 (m, 2H), 5.84–5.74 (m, 1H), 7.30–7.18 (m, 10H), 7.76 (br s, 1H); ¹³C NMR (CDCl₃): δ : 46.6, 46.7, 56.7, 57.2, 66.4, 119.4, 127.86, 128.50, 128.54, 128.73, 130.9, 135.43, 135.47, 159.7, 167.7, 170.7; MS (ESI): *m/z* = 417.3 (M+Na)⁺.

4.3.5. (4*S*,5*R*)-1,3-Dibenzyl-5-(propargyloxycarbonyl)-2-oxo-imidazolidine-4-carboxylic acid **4e**

White solid; yield: 0.190 g (97%); mp 131–135 °C; [α]_D^{25.7} = +4.5 (c 1.0, CHCl₃) [lit.^{3b} mp 132.7–135.8 °C; [α]_D²⁵ = +14.3 (c 1.0, CHCl₃)]; ee = 24%; IR (KBr): ν = 3299, 3026, 2919, 1751, 1652, 1464, 1440, 1203, 978, 704 cm⁻¹; ¹H NMR (CDCl₃): δ : 2.50 (t, 1H, *J* = 2.8 Hz), 4.12–4.06 (m, 4H), 4.66–4.56 (m, 2H), 5.00 (d, 1H, *J* = 14.8 Hz), 5.08 (d, 1H, *J* = 14.8 Hz), 6.62 (br s, 1H), 7.33–7.20 (m, 10H); ¹³C NMR (CDCl₃): δ : 46.78, 46.85, 53.1, 56.51, 57.04, 75.97, 76.43, 128.00, 128.02, 128.60, 128.67, 128.84, 128.87, 135.36, 135.48, 159.5, 167.3, 171.5; MS (ESI): *m/z* = 393.3 (M+H)⁺.

4.3.6. (4*S*,5*R*)-1,3-Dibenzyl-5-(benzyloxycarbonyl)-2-oxo-imidazolidine-4-carboxylic acid **4f**

White solid; yield: 0.215 g (97%); mp 57–59 °C; [α]_D^{25.3} = +11.5 (c 1.0, CHCl₃) [lit.^{3b} mp 57.7–60.6 °C; [α]_D²⁵ = +12.2 (c 1.0, CHCl₃)]; ee = 64%; IR (KBr): ν = 3031, 2945, 1752, 1664, 1454, 1236, 1201,

967, 741, 700 cm⁻¹; ¹H NMR (CDCl₃): δ : 3.70 (br s, 1H), 4.11–4.00 (m, 4H), 5.12–4.97 (m, 4H), 7.34–7.10 (m, 15H); ¹³C NMR (CDCl₃): δ : 46.79, 46.80, 56.5, 57.2, 67.7, 127.89, 127.94, 128.58, 128.61, 128.63, 128.70, 128.76, 128.83, 135.53, 135.65, 159.5, 167.8, 170.8; MS (ESI): *m/z* = 467.2 (M+Na)⁺.

4.3.7. (4*S*,5*R*)-1,3-Dibenzyl-5-(cyclohexyloxycarbonyl)-2-oxo-imidazolidine-4-carboxylic acid **4g**

Yellow oil; yield: 0.209 g (96%); ee = –6%; IR (KBr): ν = 3462, 2933, 2518, 1649, 1453, 1228, 765, 741, 700 cm⁻¹; ¹H NMR (CDCl₃): δ : 1.13–1.32 (m, 5H), 1.52–1.55 (m, 1H), 1.70–1.72 (m, 2H), 1.87–1.90 (m, 2H), 3.61–3.66 (m, 1H), 3.99–4.08 (m, 4H), 5.04 (d, 2H, *J* = 15.2 Hz), 7.20–7.32 (m, 10H); ¹³C NMR (CDCl₃): δ : 8.8, 21.7, 25.9, 39.4, 46.3, 52.1, 58.2, 62.5, 68.7, 127.6, 137.2, 160.2, 168.8; MS (ESI): *m/z* = 459 (M+Na)⁺.

4.3.8. (4*S*,5*R*)-1,3-Dibenzyl-2-oxo-5-[(3-phenylallyloxy)-carbonyl]imidazolidine-4-carboxylic acid **4h**

White solid; yield: 0.23 g (98%); mp 120.6–122.8; [α]_D^{22.1} = +7.0 (c 1.0, CH₃OH) [lit.^{3c} mp 122.0–122.8 °C; [α]_D^{21.9} = +7.6 (c 1.0, CH₃OH)]; ee = 92%; after recrystallization from AcOEt, **4h** was obtained in 92% yield with 98% ee, IR (KBr): ν = 3449, 3028, 2930, 1751, 1711, 1664, 1450, 1357, 1238, 1196, 748, 701 cm⁻¹; ¹H NMR (CDCl₃): δ : 4.02–4.13 (m, 4H), 4.62–4.72 (m, 2H), 5.01 (d, *J* = 14.8 Hz, 1H), 5.09 (d, *J* = 14.8 Hz, 1H), 6.13–6.21 (dt, *J* = 16.0, 6.4 Hz, 1H), 6.59 (d, *J* = 16.0 Hz, 1H), 7.18–7.39 (m, 15H), 10.53 (br s, 1H); MS (ESI): *m/z* = 470.3 (M+1)⁺.

4.4. (3*aS*,6*aR*)-1,3-Dibenzyl-tetrahydro-4*H*-furo[3,4-*d*]imidazole-2,4(1*H*)-dione **5**

To a stirred solution of **4h** (4.70 g, 10 mmol) and anhydrous CaCl₂ (1.11 g, 10 mmol) in anhydrous ethanol (52 mL) was added KBH₄ (1.62 g, 30 mmol) in three portions at 0–5 °C. Stirring was continued at the same temperature for 1 h, the reaction mixture was allowed to warm up to room temperature and stirred for another 18 h at 25 °C. The resulting mixture was then concentrated and the residue was dissolved in ethyl acetate (30 mL). The solution was extracted with 10% aqueous Na₂CO₃ (3 × 30 mL) and the combined aqueous phases were neutralized with concentrated aqueous HCl and then treated with 5% aqueous HCl (37 mL) at 60 °C. After 30 min at this temperature, the solution was extracted with CH₂Cl₂ (3 × 30 mL). The combined organic phases were dried over Na₂SO₄ and concentrated in vacuo to afford **5** as a white solid; yield: 3.16 g (98%); mp 119.4–120.1 °C; [α]_D^{25.3} = +61.2 (c 2.0, CHCl₃) [lit.¹⁴ mp 120–121 °C; [α]_D²⁵ = +61.5 (c 2.0, CHCl₃)]; IR (KBr): ν = 3031, 2919, 1775, 1706, 1415, 1365, 1237, 1209, 1146, 970, 754, 700, 639, 527 cm⁻¹; ¹H NMR (CDCl₃): δ : 3.92 (d, 1H, *J* = 8.0 Hz), 4.09–4.16 (m, 3H), 4.37 (dd, 2H, *J* = 10.4 Hz, 15.2 Hz), 4.63 (d, 1H, *J* = 15.2 Hz), 5.05 (d, 1H, *J* = 15.2 Hz), 7.24–7.36 (m, 10H); ¹³C NMR (CDCl₃): δ : 45.2, 46.9, 52.4, 54.3, 70.1, 127.8, 128.0, 128.2, 128.7, 128.8, 129.0, 135.9, 136.0, 158.1, 172.7; MS (ESI): *m/z* = 345.2 (M+Na)⁺.

4.5. (3*aS*,6*aR*)-1,3-Dibenzyl-tetrahydro-4*H*-thieno[3,4-*d*]imidazole-2,4(1*H*)-dione **6**

A mixture of **5** (3.22 g, 10 mmol), sodium ethylthioxanthogenate (1.92 g, 12 mmol), and anhydrous DMA (25 mL) was stirred at 125 °C under argon atmosphere for 5 h. After cooling, the mixture was poured into H₂O (30 mL) and extracted three times with CH₂Cl₂ (3 × 30 mL). The combined organic phases were washed successively with brine (30 mL) and H₂O (30 mL), dried over Na₂SO₄, and evaporated the solvent to give the crude product which was then purified by recrystallization from *iso*-propanol to afford the pure **6** as a white solid, yield: 2.37 g (70%); mp 125.5–

126.2 °C; $[\alpha]_{\text{D}}^{25.3} = +90.1$ (c 1.0, CHCl₃) [lit.¹⁵ mp 125–126 °C; $[\alpha]_{\text{D}}^{25} = +90.2$ (c 1.0, CHCl₃)]; IR (KBr): $\nu = 3030, 2934, 2889, 1703, 1697, 1453, 1412, 1361, 1218, 1148, 1051, 997, 903, 808, 666, 647, 581, 485 \text{ cm}^{-1}$; ¹H NMR (CDCl₃): δ : 3.28 (dd, 1H, $J = 12.4, 2.0 \text{ Hz}$), 3.37 (dd, 1H, $J = 12.4, 5.6 \text{ Hz}$), 3.81 (d, 1H, $J = 8.0 \text{ Hz}$), 4.15–4.11 (m, 1H), 4.36 (d, 1H, $J = 14.8 \text{ Hz}$), 4.37 (d, 1H, $J = 15.6 \text{ Hz}$), 4.68 (d, 1H, $J = 15.6 \text{ Hz}$), 5.03 (d, 1H, $J = 14.8 \text{ Hz}$), 7.37–7.26 (m, 10H); ¹³C NMR (CDCl₃): δ : 33.0, 45.2, 46.5, 55.8, 62.1, 127.73, 127.91, 128.0, 128.66, 128.78, 128.87, 136.2, 136.4, 158.2, 203.4; MS (ESI): $m/z = 361.1$ (M+Na)⁺.

4.6. Ethyl(3*A*,4*R*,5*R*,6*A*)-5-(1,3-dibenzyl-2,3,3*a*,4,6,6*a*-hexahydro-2-oxo-1*H*-thieno[3,4-*d*]imidazol-5-hydroxyl)pentanoate 7

To a suspension of zinc powder (1.62 g, 24.8 mmol) in anhydrous DMF (8 mL) was added iodine (0.2 g, 0.8 mmol) under nitrogen atmosphere and the reaction mixture was heated to 80 °C. Ethyl 5-bromopentanoate (3.45 g, 16.5 mmol) was then added dropwise over 30 min and the resulting mixture was kept stirring at the same temperature for 4 h. Compound **6** (2 g, 5.92 mmol), 10% Pd/C (0.07 g), THF (30 mL), and toluene (40 mL) were added in one portion into the resulting mixture and the mixture was stirred at 30 °C for 30 h. The reaction was quenched by saturated aq NH₄Cl (30 mL), and the mixture was filtered after being stirred for 10 min. The filtrate was extracted by AcOEt (3 × 30 mL). The combined organic phases were washed with brine, dried over Na₂SO₄, and concentrated in vacuo. The residue was further purified by chromatography on a silica gel column (AcOEt/PE = 1/1.5) to give **7** as a yellow oil, yield: 2.08 g (75%); $[\alpha]_{\text{D}}^{20.9} = +20.2$ (c 1.0, CHCl₃) [lit.^{3c} $[\alpha]_{\text{D}}^{20} = +20.9$ (c 1.0, CHCl₃)]; IR (KBr): $\nu = 2961, 2594, 1670, 1387, 1259, 665 \text{ cm}^{-1}$; ¹H NMR (CDCl₃): δ : 1.23–1.27 (m, 3H), 1.44–1.45 (m, 2H), 1.60–1.62 (m, 2H), 2.24–2.33 (m, 4H), 2.72 (m, 2H), 3.65 (m, 1H), 3.94 (d, 1H, $J = 22.4 \text{ Hz}$), 4.02 (d, 1H, $J = 22.4 \text{ Hz}$), 4.10–4.20 (m, 4H), 4.88 (d, 1H, $J = 15.6 \text{ Hz}$), 5.10 (d, 1H, $J = 15.6 \text{ Hz}$), 7.24–7.32 (m, 10H); MS (ESI): $m/z = 469$ (M+1)⁺.

4.7. Ethyl(3*A*,4*S*,5*R*,6*A*)-5-(1,3-dibenzyl-2,3,3*a*,4,6,6*a*-hexahydro-2-oxo-1*H*-thieno[3,4-*d*]imidazol-5-yl)pentanoate 8

An ice cold solution of **7** (2.06 g, 4.4 mmol) and NaBH₄ (0.5 g, 13.2 mmol) in dry CH₂Cl₂ (25 mL) was treated dropwise with trifluoroacetic acid (3 g, 26.4 mmol) over 30 min. The reaction mixture was then allowed to stir at 25 °C for 12 h under argon atmosphere. After adding the saturated aq NaHCO₃, the mixture was extracted with CH₂Cl₂ (3 × 30 mL). The combined organic phases were washed with brine, dried over Na₂SO₄, and concentrated in vacuo. The residue was further purified by chromatography on a silica gel column (AcOEt/PE = 1/1.5) to give **8** as a colorless oil, yield: 1.81 g (91%); $[\alpha]_{\text{D}}^{22.1} = -24.1$ (c 1.0, CH₃OH) [lit.^{3c} $[\alpha]_{\text{D}}^{21.9} = -24.9$ (c 1.0, CH₃OH)]; IR (KBr): $\nu = 1728, 1678, 1453, 1243, 700 \text{ cm}^{-1}$; ¹H NMR (CDCl₃): δ : 1.25–1.28 (t, 3H, $J = 8 \text{ Hz}$), 1.57–1.65 (m, 6H), 2.27–2.31 (t, 2H, $J = 7.2 \text{ Hz}$), 2.68, 2.72 (dddd, 2H, $J = 5.6 \text{ Hz}, J = 5.6 \text{ Hz}$), 3.05 (m, 1H), 3.86–4.12 (m, 6H), 4.72 (d, 1H, $J = 15.2 \text{ Hz}$), 5.05 (d, 1H, $J = 15.2 \text{ Hz}$), 7.23–7.34 (m, 10H); ¹³C NMR (CDCl₃): δ : 14.3, 24.7, 28.5, 28.7, 34.2, 34.8, 46.6, 48.0, 54.3, 60.3, 61.2, 62.6, 127.3, 128.3, 128.7, 136.9, 137.0, 161.1, 173.6; MS (ESI): $m/z = 453.3$ (M+1)⁺.

4.8. (+)-Biotin 1

A mixture of **8** (4.52 g, 10 mmol) and 47% aq HBr (40 mL) was heated to reflux with vigorous stirring. After 48 h the reaction mixture was extracted with toluene (3 × 30 mL) and the aqueous phase was separated and concentrated in vacuo. The residue was dissolved in H₂O (18 mL). With vigorous stirring, the solution of trichloromethyl chloroformate (8.91 g, 45 mmol) in anisole (40 mL)

was added dropwise over 2 h and the resulting mixture was allowed to stir at 35 °C overnight. The pH value of the reaction mixture was maintained at 10 by addition of 40% aqueous KOH. The separated aqueous phase was acidified to pH 1 with concentrated aqueous HCl and cooled with ice water. The crystals that formed were collected by filtration and recrystallized from H₂O to give pure **1** as a white crystalline powder, yield: 2.01 g (82%); mp 232.1–233.0 °C; $[\alpha]_{\text{D}}^{22.3} = +91.3$ (c 1.0, 0.1 N NaOH) [lit.¹⁶ mp 232–233 °C; $[\alpha]_{\text{D}}^{21.9} = +91.2$ (c 1.0, 0.1 N NaOH)]; IR (KBr): $\nu = 3359, 3308, 2961, 2469, 1941, 1707, 1480, 1318, 1270, 1154, 1015, 842, 753, 651 \text{ cm}^{-1}$; ¹H NMR (DMSO): δ : 1.37–1.30 (m, 2H), 1.62–1.41 (m, 4H), 2.20 (t, 2H, $J = 7.6 \text{ Hz}$), 2.57 (d, 1H, $J = 12.4 \text{ Hz}$), 2.82 (dd, 1H, $J = 12.4, 5.2 \text{ Hz}$), 3.12–3.07 (m, 1H), 4.15–4.11 (m, 1H), 4.30 (dd, 1H, $J = 7.6, 5.2 \text{ Hz}$), 6.33 (s, 1H), 6.40 (s, 1H), 11.9 (s, 1H); ¹³C NMR (CDCl₃): δ : 25.0, 28.52, 28.57, 33.9, 40.0, 55.8, 59.6, 61.5, 163.1, 174.8; MS (ESI): $m/z = 267$ (M+Na)⁺.

Acknowledgment

This work was supported by the DSM Nutritional Products Ltd in Switzerland.

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